Paleomagnetism of Upper Precambrian Sedimentary Rocks From the Desert Range, Nevada

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About 300 oriented samples were obtained from upper Precambrian and possibly lowest Cambrian strata in a homoclinal sequence conformable with Lower Paleozoic strata in the Desert Range, Nevada. These samples are from miogeoclinal sedimentary rocks of the uppermost Johnnie Formation (upper Precambrian) and overlying Stirling Quartzite (uppermost Precambrian and Lower(?) Cambrian). For about 100 of the samples, thermal demagnetization revealed a characteristic magnetization with a low inclination. The characteristic magnetization of the uppermost Johnnie has two polarities, which allows its direction to be established despite incomplete removal of a secondary component of recent origin. Petrographic and stratigraphic evidence suggest that this characteristic magnetization was acquired penecontemporaneously with deposition. The age of the magnetization probably lies in the range 675–575 m.y. B.P., based on estimates of sedimentation rates in the Desert Range and on proposed lithologic correlations and isotopic age control elsewhere in the Cordilleran Geosyncline. A fine-grained, red-purple unit in the middle Stirling Quartzite exhibits a single-polarity (reversed) characteristic magnetization with a direction roughly like the directions of the uppermost Johnnie or the Lower Cambrian Wood Canyon Formation, which conformably overlies the Stirling. The paleomagnetic pole from the uppermost Johnnie is about 47° from the pole for the Wood Canyon. It seems probable therefore that at least 45° of apparent polar wandering occurred with respect to North America in latest Precambrian time.

INTRODUCTION

In the companion paper [*Gillett and Van Alstine*, 1979] we described the motivation for and the methodology of a systematic paleomagnetic investigation of upper Precambrian through Lower Paleozoic miogeoclinal strata in the Desert Range, Nevada. In that paper we presented paleomagnetic data from the Lower and Middle Cambrian parts of this section. Beneath these Cambrian beds lie nearly 3000 m of apparently conformable strata, which comprise the Johnnie Formation (upper Precambrian) and Stirling Quartzite (uppermost Precambrian and possibly Lower Cambrian). In this paper we present paleomagnetic results from the uppermost Johnnie Formation and Stirling Quartzite, in order to extend the North American apparent polar wander path into latest Precambrian time and to investigate the polarity history of the geomagnetic field near the Precambrian-Cambrian boundary.

UPPERMOST JOHNNIE FORMATION (RAINSTORM MEMBER)

The Johnnie Formation (upper Precambrian) of the southern Great Basin is a heterogeneous sequence of shale, siltstone, quartzite, limestone, and dolomite. In the Desert Range, where the Johnnie attains its maximum known thickness of more than 1500 m, *Stewart* [1970] has recognized five members, the uppermost of which is the Rainstorm Member (Figure 1). At a locality about 5 km northwest of that sampled in this study, *Stewart and Barnes* [1966] and *Stewart* [1970] have measured and described three subdivisions of the Rainstorm Member. In ascending stratigraphic order they are (1) a siltstone unit, 23 m thick, consisting predominantly of grayish olive to purple siltstone and including a distinctive, 1.8-m-thick oolitic dolomite (the 'Johnnie oolite'), (2) a carbonate unit, 198 m thick, consisting of generally pale red to grayish red intergradational limestone and siltstone, containing ripple marks, drag marks, and flute casts, and (3) a siltstone and quartzite unit, 64 m thick, consisting of grayish olive siltstone and pale yellowish brown quartzite. The stratigraphic thicknesses of the three units of the Rainstorm at the locality sampled in this study are about 20% less than those reported at *Stewart*'s [1970] Desert Range locality 70.

The presence of some minor faults and the incomplete exposure of some slope-forming beds necessitated our sampling the Rainstorm Member in four stratigraphically overlapping sections (G, H, I, and J in Figure 1 of *Gillett and Van Alstine* [1979]). Eighty-seven oriented samples were obtained from the Rainstorm Member, with the following stratigraphic distribution: 8 samples from the siltstone unit (6 samples of purple siltstone and 2 samples of the Johnnie oolite); 74 samples from the carbonate unit; and 5 samples from the siltstone and quartzite unit. These uppermost five samples behaved unstably upon thermal demagnetization so that not even their polarity could be ascertained; hence they will not be discussed further. Results from the lower two units within the Rainstorm Member are presented below.

The six samples from the siltstone unit of the Rainstorm Member were progressively demagnetized at six steps between 400° and 640°C. Directions of natural remanent magnetization (NRM) of most of these samples were displaced from the direction of the present axial dipole field. At progressively higher temperatures, some of the directions became more westerly, and some more easterly, although they did not become antiparallel. This demagnetization behavior suggests that in addition to a component aligned roughly with the direction of the present axial field, the siltstone unit of the Rainstorm contains a two-polarity magnetization with a much different direction. This remanence, which represents the characteristic magnetization of these rocks, appears to have a shallow inclination and either easterly or westerly declination. The characteristic magnetization directions cannot be pre-

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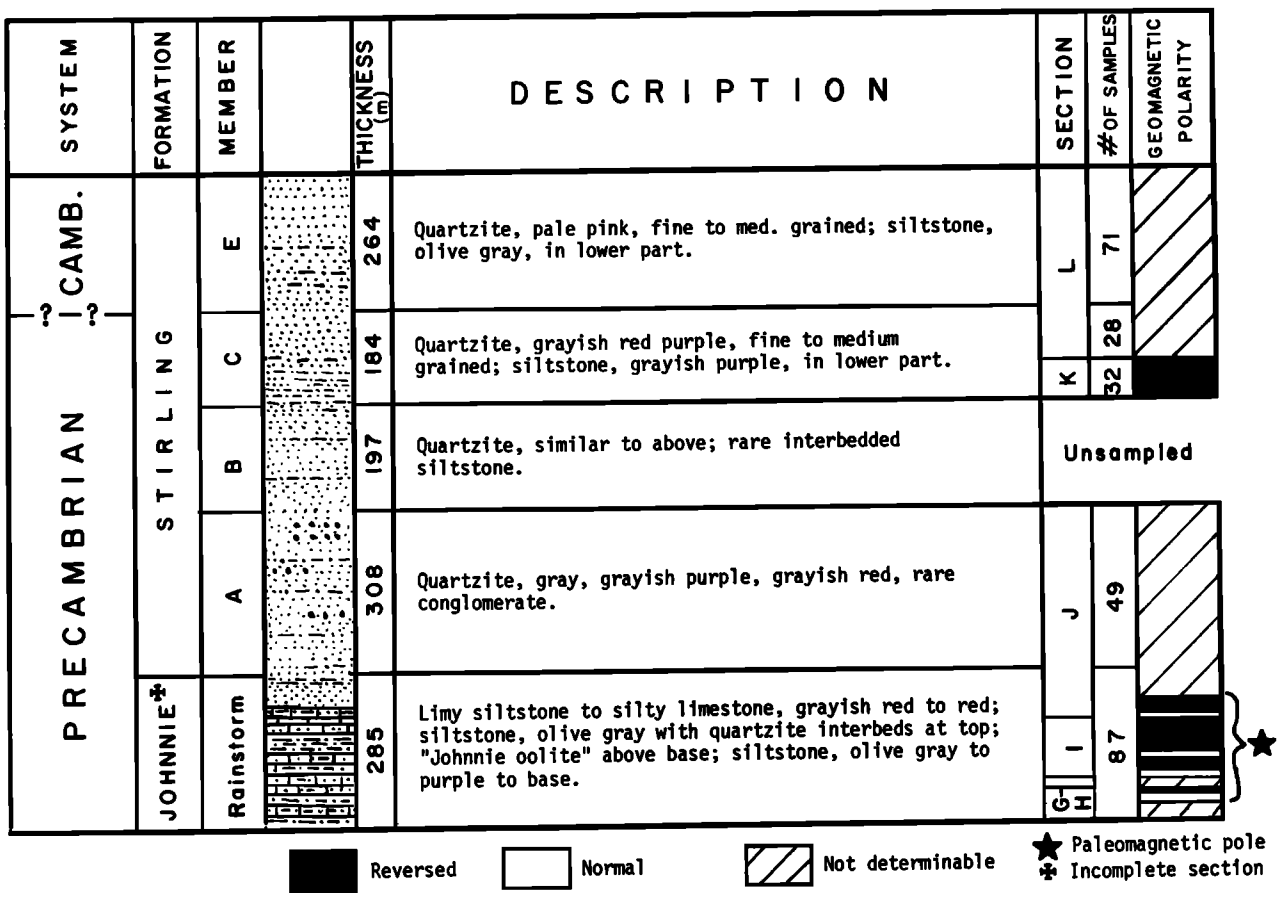


Fig. 1. Chart showing description of strata, stratigraphic distribution of samples, and inferred geomagnetic polarity for upper Precambrian and Lower(?) Cambrian formations sampled in this study. Descriptions adapted from *Stewart and Barnes* [1966]. The beds at the top of this figure are conformable with those at the bottom of Figure 2 of *Gillett and Van Alstine* [1979].

cisely defined, however, because a significant part of the present axial field component has a blocking temperature at least as high as that of the characteristic magnetization.

In contrast to the siltstone samples the two samples from the Johnnie oolite had NRM directions roughly aligned with the direction of the present axial field. Upon progressive demagnetization beginning at 200°C, the samples lost most of their intensity of magnetization by 300°C, and the directions of magnetization changed erratically at higher temperatures. Hence not even the polarity of the Johnnie oolite could be ascertained. Examination of a thin section from the Johnnie oolite shows it to be an ooid grainstone, with voids commonly remaining between the grains. Rusty-colored stylolitic veins are common, and a rusty color pervades the entire rock. This

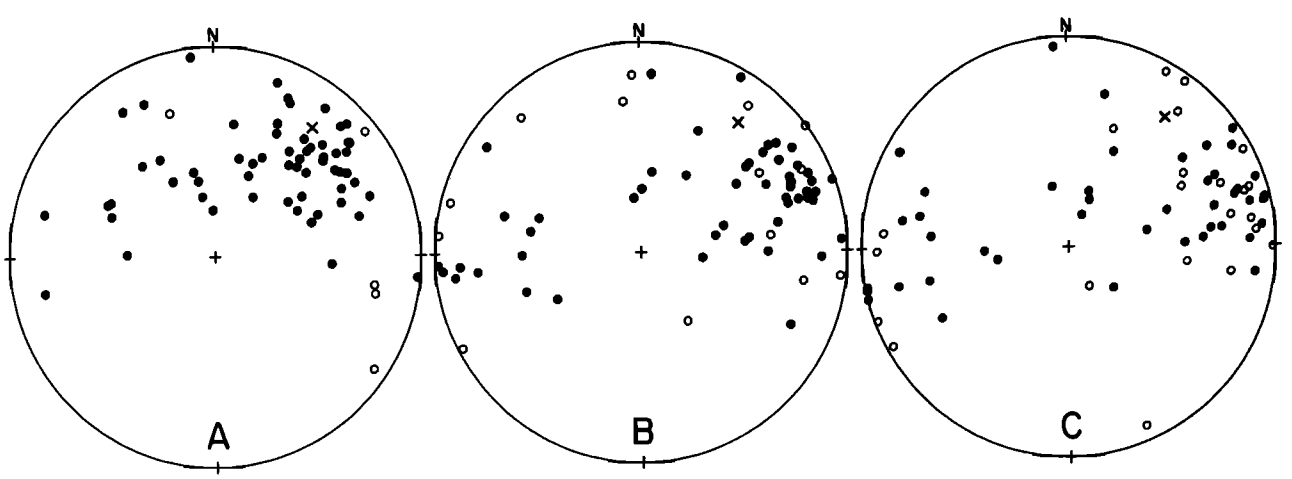


Fig. 2. Directions of magnetization at selected steps from progressive thermal demagnetization of samples from the carbonate unit of the Rainstorm Member of the Johnnie Formation. Open (solid) circles are on upper (lower) hemisphere, respectively. Cross marks the direction of the present axial dipole field at the sampling site after rotation for the mean attitude of beds. The directions from the stratigraphically highest 10 samples are not plotted, because they are from poorly exposed, slightly rotated blocks. (*a*) NRM; (*b*) 550°C; and (*c*) 650°C.

TABLE 1. Late Precambrian Paleomagnetic Pole From the Desert Range, Nevada

Formation	Section*	Mean Attitude		$\bar{J}_{mode}$, A m²/kg × 10^{-7}	$\bar{J}_{mode}/\bar{J}_{NRM}$	Geographic Coordinates, deg		Stratigraphic Coordinates, deg		β_{95} (N_{mode}/N_{tot})	Apparent North Pole	Corrected North Pole
		Strike	Dip			D	I	D	I			
Johnnie† (normal)	H, I, J	344	51°E	6.6	0.50	265	−41	261	+7	11° (10/20)		
Johnnie† (reversed)	H, I, J	344	51°E	5.1	0.31	73	+62	74	+8	7° (31/45)		
Johnnie† (average)	H, I, J	344	51°E					78	+1		10°S, 162°E	37°S, 188°E

$\bar{J}_{mode}$ is the geometric mean of the specific intensities of magnetization at the demagnetization step upon which the mode is based (i.e., 650°–660°C). Geographic coordinates are the mode of specimen directions uncorrected for the bedding attitude. Stratigraphic coordinates are the mode of specimen directions after correction for local bedding attitudes, not for the mean attitude of the entire formation. β_{95} is an estimate of the 95% confidence limits for the mode. The statistic β_{95} is derived from the largest subset (N_{mode}) of the total sample (N_{tot}) that has a mean identical with the mode of the total sample. This statistic is defined as the Fisherian half angle of the cone of 95% confidence for the mean of this subset. The apparent north paleomagnetic pole has been calculated from the average stratigraphic coordinates before correction for the 36° clockwise rotation with respect to the Colorado Plateau (see text). The corrected north paleomagnetic pole has been calculated from the average stratigraphic coordinates after correction for the 36° clockwise rotation with respect to the Colorado Plateau (see text).

*Refers to paleomagnetic section locations of Figure 1 of *Gillett and Van Alstine* [1979].

†Directions from the carbonate unit of the Rainstorm Member of the (uppermost) Johnnie Formation. Normal refers to normal polarity; reversed refers to reversed polarity; average refers to average of the mode of normal-polarity directions with the mode of reversed-polarity directions (after first rotating the normal-polarity mode through 180°).

rusty material, which is probably iron bearing and of recent origin, in all likelihood dominates the NRM and the magnetic behavior at low-temperature demagnetization steps.

About half of the 74 samples from the carbonate unit of the Rainstorm Member were progressively demagnetized at 10 steps between 200° and 670°C; the rest were demagnetized at five steps between 550° and 670°C. The NRM directions of samples from this unit were scattered about the direction of the present axial field (Figure 2). At successively higher temperatures, directions from most samples migrated away from the present axial field direction to form bimodal distributions. The modes have low, positive inclinations and declinations in the northeast and southwest quadrants. By 650°C the intensity of magnetization of most samples had decreased to 45–75% of their NRM values, and the directions had become nearly antiparallel. At 670°C the directions from most samples abruptly scattered.

Twelve duplicate specimens from the Rainstorm Member were subjected to progressive chemical demagnetization by soaking in concentrated HCl up to a maximum of 500 hours. The specimens were ring shaped to increase the surface area for leaching; they were made by drilling holes about 1.5 cm across in wafers about 7 mm thick. Chemical demagnetization resulted in incomplete separation of magnetic components. A distribution somewhat skewed toward the positions revealed by thermal demagnetization could be discerned after prolonged immersion, and the intensity of magnetization decreased by about 50%. However, scatter of directions remained high, the directions of some specimens changed rather erratically, and many directions remained near the direction of the present axial field. Moreover, the abundance of carbonate in some samples caused them to disintegrate.

Six thin sections from the carbonate unit of the Rainstorm Member were examined in transmitted and reflected light. The rocks consist predominantly of silt-sized (20–30 μm) quartz grains, with subordinate feldspar and rare detrital white mica. Vague lamination defined by grain-size variations is present. The carbonate cement that occurs in these rocks is micritic and probably is calcite, as it reacts readily with cold, dilute HCl. The rocks are pervasively tinted a light pinkish orange. Local reddish or red-orange hematitic patches and aggregates occur; these probably were formed by alteration of iron-bearing precursors. Minute (~1 μm) flecks of specular hematite are occasionally associated with the aggregates; these flecks presumably also are authigenic, in contrast to the probably detrital grains described below.

Discrete grains of specular hematite, ~10 μm across, constitute about 1% of these rocks. Many of them have squarish outlines, suggesting they may be martite. Most of these grains are probably detrital, as suggested by the following observations: (1) The grains are generally concentrated in lenses, stringers, and laminae that apparently are heavy mineral trains. (2) The grains commonly show rounding, suggestive of transport. (3) The grains have crisp contacts with the surrounding rock.

Modes representing each of the two polarities of the characteristic magnetization of the Rainstorm Member were determined from the 650°C demagnetization step of the carbonate unit (Figure 2 and Table 1). These modes are not antiparallel, and the distributions of directions are skewed. Evidently, the present axial field component has not been completely removed, even by 650°C. An estimate of the direction representing one polarity of the characteristic magnetization can be made by rotating one of the modes through 180° and then averaging it with the mode of opposite polarity [cf. *McElhinny*, 1973, p. 85]. This procedure yields a direction with declination D = 78° and inclination I = +1° in stratigraphic (i.e., dip corrected) coordinates for the characteristic magnetization with easterly declination. This direction is assumed to represent reversed polarity, because of its proximity to the reversed-polarity direction from the middle member of the Lower Cambrian Wood Canyon Formation [*Gillett and Van Alstine*, 1979].

An apparent polarity zonation is observed in the carbonate unit of the Rainstorm Member (Figure 3). The carbonate unit appears to contain at least 10 polarity zones, each of which is represented by at least two samples and is therefore locally reproducible.

The high blocking temperature (>650°C) of the characteristic magnetization of the carbonate unit of the Rainstorm

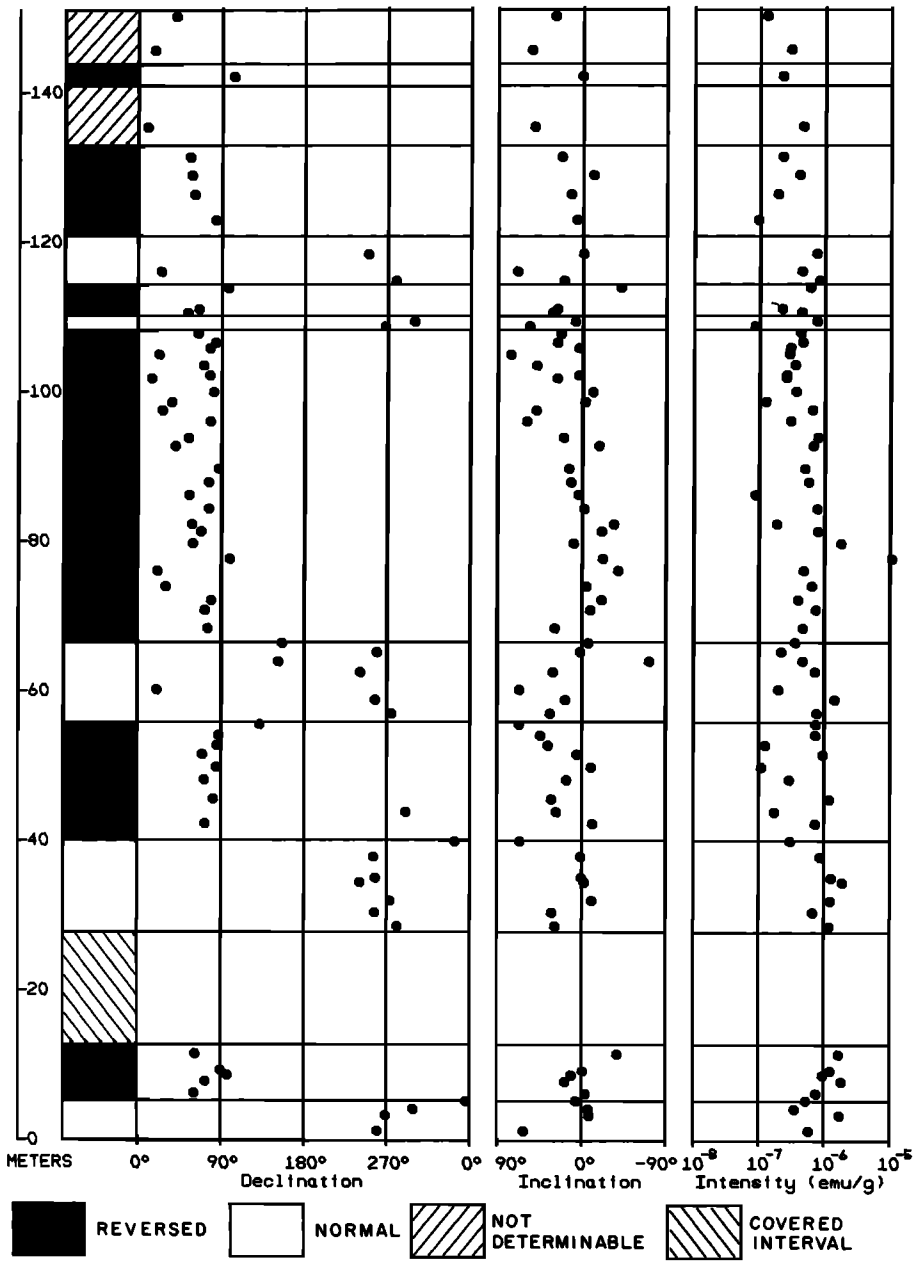


Fig. 3. Apparent polarity zonation of the carbonate unit of the Rainstorm Member of the Johnnie Formation in the Desert Range, Nevada. Polarity assignments have been made by inspection of vector demagnetization diagrams of each sample. The base of the section is the top of the Johnnie oolite; the top of the section is the top of the carbonate unit (1 emu/g = 1 A m^2/kg).

Member demonstrates that this magnetization resides in essentially pure hematite. Whether this magnetization is a detrital remanent magnetization (DRM) or a chemical remanent magnetization (CRM) is ambiguous. The thin-section observations, which show common, apparently detrital specular hematite in these rocks, suggest that the specular hematite might contain a significant fraction of the characteristic magnetization. In this case the magnetization could be a DRM, acquired contemporaneously or penecontemporaneously with deposition. However, authigenic hematite also occurs in these rocks and could contribute a component of CRM. Thermal demagnetization experiments cannot distinguish conclusively between remanences residing in authigenic versus detrital hematite. Chemical demagnetization (which, in favorable cases, can distinguish CRM from DRM by preferentially leaching away fine-grained authigenic hematite) gave equivocal results for rocks from the Rainstorm Member.

Even if the characteristic magnetization were largely a CRM, stratigraphic evidence suggests that it was acquired in the late Precambrian. The Rainstorm Member, unlike the underlying strata of the Johnnie Formation, contains distinctive units that can be correlated in detail over a wide area. In particular, the Johnnie oolite and the reddish middle carbonate unit are present throughout the southern Great Basin [*Stewart*, 1970]. Moreover, hematite-bearing beds that can be confidently correlated on lithologic grounds with the Rainstorm Member occur in late Precambrian strata exposed several hundred kilometers to the north, in eastern Nevada and western Utah [*Stewart*, 1974]. If the hematite of the carbonate unit of the Rainstorm had been produced chiefly by late altera-

tion in the Phanerozoic, its distribution would probably be more spotty than is suggested by the regional stratigraphy.

As discussed in the companion paper, deposition in this part of the miogeocline was essentially continuous throughout the late Precambrian and Paleozoic. By the late Paleozoic the Rainstorm was buried to depths of about 7 km, far below surface weathering and oxidation. Furthermore, the Rainstorm is sandwiched between hundreds of meters of quartzite, siltstone, and shale. Because these rocks are not very permeable [*Winograd and Thordarson*, 1968], it seems implausible that oxidizing groundwaters could penetrate them to oxidize the Rainstorm Member on a regional scale in the Phanerozoic. The high blocking temperature (>650°C) rules out the possibility that the characteristic magnetization of the Rainstorm could be a viscous partial thermoremanent magnetization associated with heating during burial [cf. *Pullaiah et al.*, 1975].

Another indication of a late Precambrian time of acquisition of the characteristic magnetization of the Rainstorm Member comes from consideration of its corresponding paleomagnetic pole position. A paleomagnetic pole for the Rainstorm computed directly from its characteristic magnetization direction (in stratigraphic coordinates) lies more than 60° from the post-Early-Ordovician apparent polar wander (APW) path for North America (Figure 4). This suggests that the characteristic magnetization of the Rainstorm predates the Ordovician. As discussed in the companion paper, the pole during the Early and early Middle Cambrian seems to have been relatively stationary with respect to North America, near the position of the paleomagnetic pole from the Lower and Middle Cambrian Tapeats Sandstone of the Colorado Plateau [*Elston and Bressler*, 1977]. Even the pole from the Tapeats, however, is about 15° from that determined directly from the characteristic magnetization direction of the Rainstorm Member (Figure 4). This suggests that the magnetization of the Rainstorm was acquired prior to the Early Cambrian.

In the companion paper we showed that characteristic magnetization directions derived from Lower and Middle Cambrian strata of the Desert Range were consistent with a net 36°

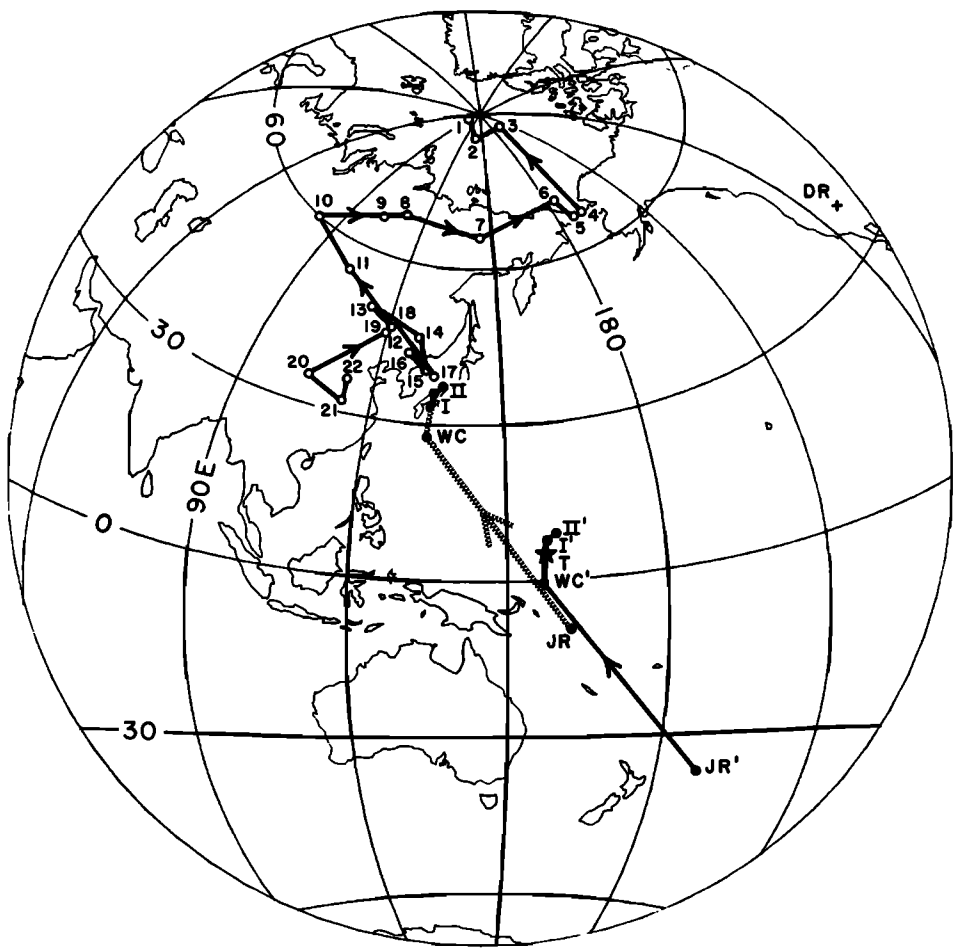


Fig. 4. Upper Precambrian and Cambrian paleomagnetic poles from the Desert Range before (unprimed symbols) and after (primed symbols) correction for the proposed 36° clockwise vertical-axis rotation of the Desert Range section with respect to the Colorado Plateau [*Gillett and Van Alstine*, 1979]. II, uppermost Carrara and lowermost Bonanza King Formations (Middle Cambrian); I, remainder of the Carrara Formation (Lower and Middle Cambrian); and WC, Wood Canyon Formation (Lower Cambrian), all from *Gillett and Van Alstine* [1979]; JR, Rainstorm Member of the Johnnie Formation (upper Precambrian), from this study. The star labeled 'T' is the pole from the Tapeats Sandstone (Lower and Middle Cambrian) of the Colorado Plateau [*Elston and Bressler*, 1977]. The cross labeled 'DR' marks the sampling site. Also shown is the post-Early-Ordovician part of the apparent polar wander path for North America determined by *Van Alstine* [1979]. Numbers on the path refer to nearly uniform, 22-m.y. time intervals: 1–3, Cenozoic; 4–7, Cretaceous; 7–9 Jurassic; 10–11, Triassic; 12–13, Permian; 14–15, Pennsylvanian; 16–17, Mississippian; 18–19, Devonian; 20–21, Silurian; and 21–22, Ordovician.

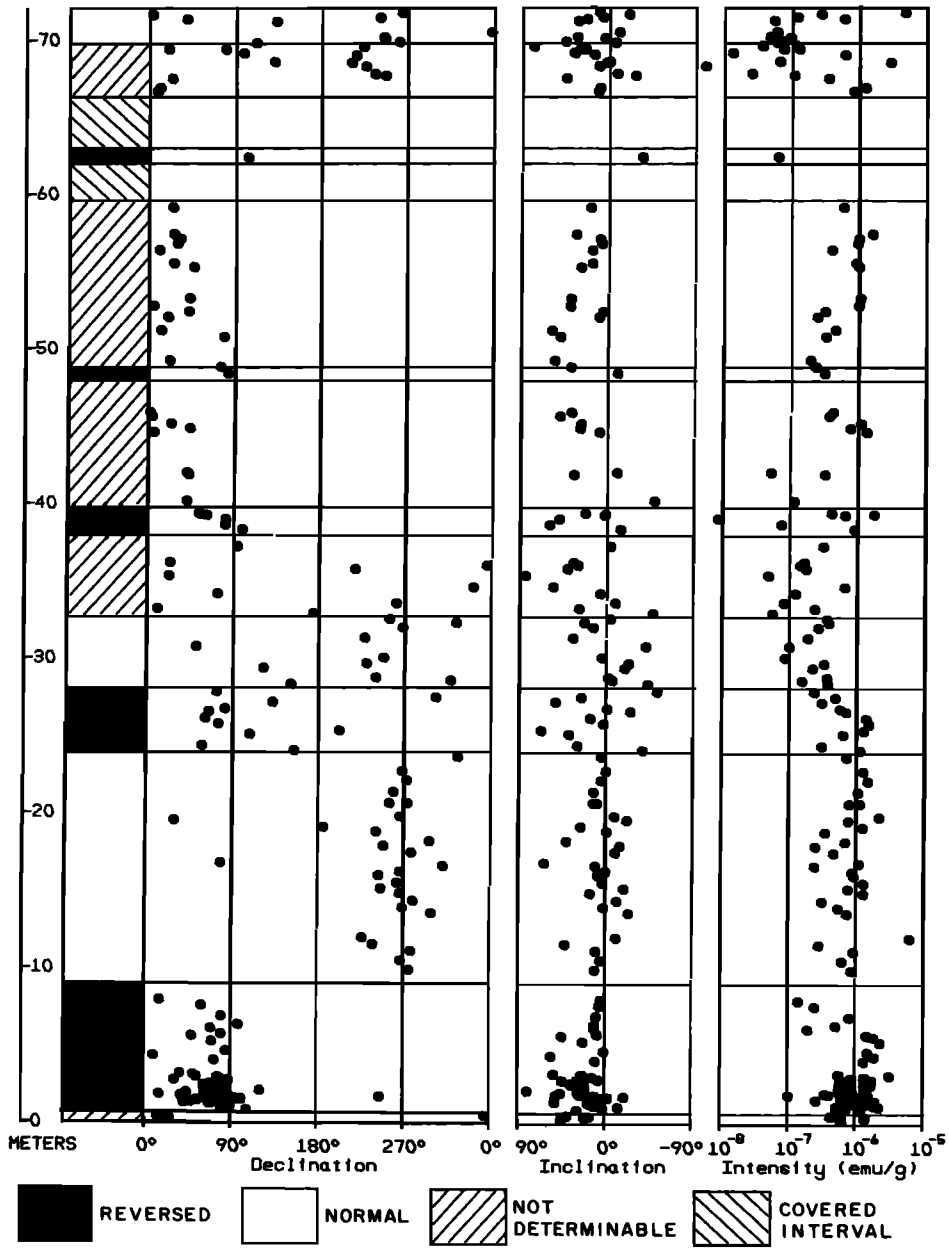


Fig. 5. Apparent polarity zonation of the carbonate unit of the Rainstorm Member of the Johnnie Formation in the Nopah Range, California. Polarity assignments have been made by inspection of vector demagnetization diagrams of each sample. The base of the section is the top of the Johnnie oolite; the top of the section is the top of the carbonate unit (1 emu/g = 1 A m^2/kg).

clockwise rotation of this part of the range. This vertical-axis rotation was inferred from the mismatch in paleomagnetic declination between the Cambrian strata from the Desert Range and the partly coeval Tapeats Sandstone. The net rotation was attributed chiefly to oroflexural bending in the late Cenozoic but may in part have been caused by thrusting in the Mesozoic. Because the Rainstorm Member was sampled in the same homocline as the Cambrian strata, a 36° declination correction must also be applied to the characteristic magnetization of the Rainstorm. The corrected north paleomagnetic pole from the Rainstorm (37°S, 188°E; Table 1) falls even farther from both the post-Early-Ordovician APW path for North America and from the pole for the Tapeats (Figure 4). This lack of correspondence between the paleomagnetic pole for the Rainstorm and Phanerozoic poles suggests that the characteristic magnetization of the Rainstorm was indeed acquired in the late Precambrian.

The regional extent of the distinctive units in the Rainstorm Member and the the polarity reversals found in the Desert Range suggest yet another test for the time of acquisition of its characteristic magnetization. If the magnetization were acquired soon after deposition, and if deposition rates were relatively uniform, the distinctive polarity zonation of the Rainstorm in the Desert Range would be reproducible at other localities. As an initial step in such a test, the authors and colleagues have sampled the Rainstorm in the southern Nopah Range, about 120 km southwest of the Desert Range section. The data obtained to date suggest that the polarity zonation of the Rainstorm in the Nopah Range is indeed similar to that in the Desert Range (Figure 5).

STIRLING QUARTZITE

The Stirling Quartzite, which conformably overlies the Rainstorm Member of the Johnnie Formation, consists of light-colored quartzite with minor siltstone and dolomite (Figure 1). Because it lies below the earliest occurrence of an olenellid trilobite/archeocyathid fauna, the Stirling has been assigned a late Precambrian age [*Stewart*, 1970]. However, *Nelson* [1978] has suggested that the Precambrian-Cambrian boundary may lie in the upper Stirling, on the basis of proposed lithologic correlations with more fossiliferous strata in the White-Inyo Mountains of eastern California. The Stirling has been divided into five informal members of wide extent, designated A, B, C, D, and E by *Stewart* [1970]. The dolomitic D member is absent in the Desert Range, as it is in most of the more eastern sections of the miogeoclinal sequence [*Stewart*, 1970]. At a locality 6.5 km south of the Desert Range section sampled in this study, *Stewart and Barnes* [1966] and *Stewart* [1970] have measured and described a total of 950 m of beds belonging to the A, B, C, and E members.

Of a total of 180 samples collected from the Stirling, 142 are from light-colored quartzite, and 6 are from interbeds of dolomitic sandstone and purple siltstone from the A, C, and E members. The other 32 samples are from purple to red-purple siltstones and claystones from near the base of the C member. The siltstones and claystones from the C member are the only rocks from the Stirling in which a characteristic magnetization could be discerned.

All samples of the siltstone and claystone were subjected to progressive thermal demagnetization at six steps between 400° and 640°C. The NRM directions of most samples clustered near the direction of the present axial field (Figure 6). By 500°C, directions from about half of the samples had remained essentially stationary, whereas the other half had moved slightly away from the present axial field toward a direction with more easterly declination and shallower inclination. Bimodal distributions of directions were evident at the 610° and 625°C demagnetization steps. At 640°C, however, samples representing each of the two modes displayed a viscous magnetization; hence demagnetization of these samples at higher temperatures was not attempted.

These progressive thermal demagnetization experiments reveal the existence of two components of magnetization in about half these rocks: (1) a component of recent origin aligned roughly with the direction of the present axial field and (2) a characteristic component, with a more shallow inclination and an easterly declination. The characteristic magnetization has only one polarity. Because the stability spectra of the two components overlap, the directions from samples in which both components are present never reach a stable endpoint. This, together with the single-polarity nature of the characteristic magnetization, precludes a precise determination of its direction. However, this magnetization appears to have a direction similar to the reversed-polarity directions from the Rainstorm Member of the upper Precambrian Johnnie Formation ($D = 78°$, $I = +1°$) or the middle member of the Lower Cambrian Wood Canyon Formation ($D = 124°$, $I = -3°$) [*Gillett and Van Alstine*, 1979]. This suggests that the characteristic magnetization in the C member of the Stirling was acquired in the Precambrian and is of reversed polarity.

All samples of red-purple siltstone and claystone were examined under a low-power binocular microscope; in addition, two thin sections of red-purple claystone were examined in transmitted and reflected light. The two rock types are intergradational in lithology, and the variation in grain size is not correlated with the thermal demagnetization behavior. An opaque, red hematitic cement is abundant in the fine-grained parts of these rocks, accounting for their color in hand specimen (5 P 4/2, purple, to 5 RP 4/2, red-purple). This cement is the dominant form of hematite in the rocks. Specular hematite was found only in one of the two thin sections, and some of this is probably authigenic, as suggested by its association with hematitic aggregates.

The hematitic cement probably carries the major part of the characteristic magnetization in the siltstones and claystones from the Stirling Quartzite. Although this cement must postdate deposition to some degree, textural evidence suggests that it was formed quickly, implying that any CRM associated with the cement was probably acquired penecontemporaneously with deposition. These textural relations are similar to those in red-purple mudstones in the middle member of the overlying Wood Canyon Formation [*Gillett and Van Alstine*, 1979]. The siltstones and claystones contain laminae; lenses; and irregular, crosscutting, infilled cracks or burrows, all of which are composed of coarser, quartzofeldspathic material. The cement tends to be confined to the fine-grained areas, so that the coarser areas remain light colored and the contact of the two is commonly sharp. This suggests that at least the precursor to the cement was present shortly after deposition. One would expect hematite from late alteration (1) to show more diffuse contacts and (2) to follow preferentially the coarse-grained areas because of their initially higher permeability.

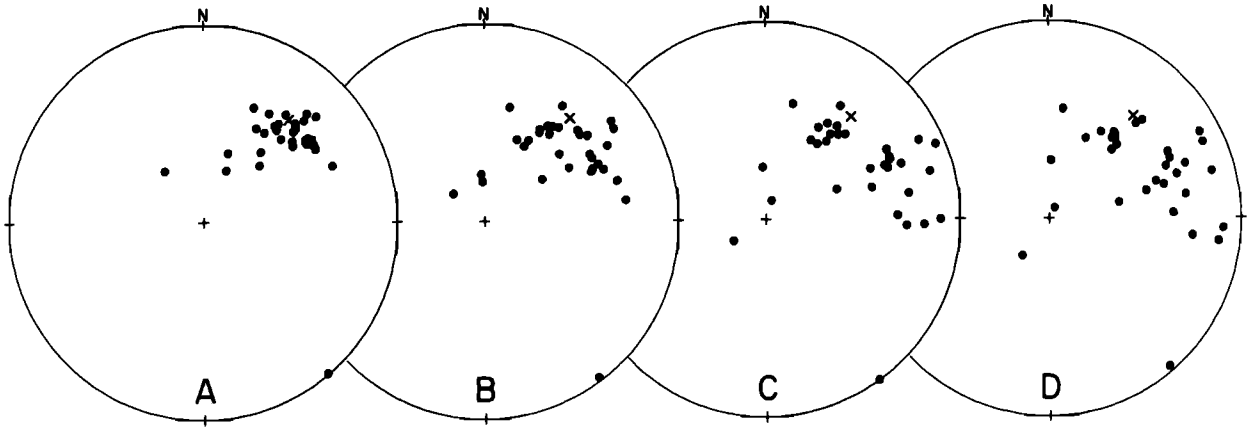


Fig. 6. Directions of magnetization at selected steps from progressive thermal demagnetization of red-purple siltstones and claystones from the lower C member of the Stirling Quartzite. Symbols are as in Figure 2. (*a*) NRM; (*b*) 500°C; (*c*) 610°C; and (*d*) 640°C.

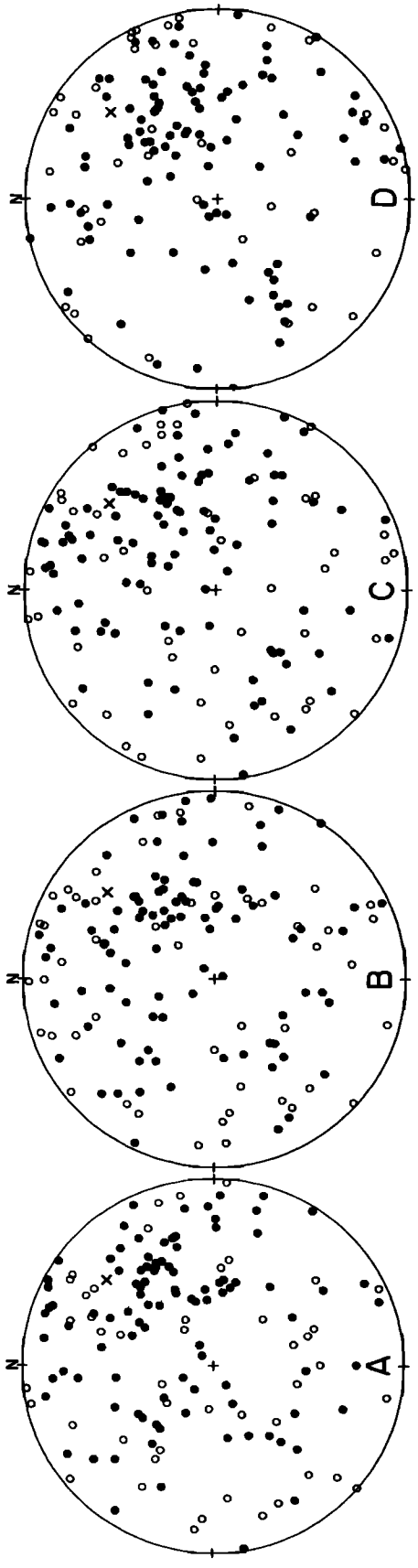


Fig. 7. Directions of magnetization at selected steps from progressive thermal demagnetization of quartzite samples from the A, C, and E members of the Stirling Quartzite. Symbols are as in Figure 2. (*a*) NRM; (*b*) 400°C; (*c*) 550°C; and (*d*) 640°C.

Progressive thermal demagnetization did not reveal a direction of characteristic magnetization from other units of the Stirling Quartzite. The quartzitic and dolomitic samples from the Stirling were progressively thermally demagnetized at five steps between 400° and 640°C; about half were initially demagnetized at 200° and 300°C as well. Directions from the quartzites and from four brown dolomite interbeds were poorly grouped before demagnetization, and they remained dispersed at every demagnetization step (Figure 7). Directions from some of these samples, especially the dolomitic interbeds, changed through large angles in an erratic manner. However, directions from most samples, which were predominantly quartzite, were extremely stable to thermal demagnetization even by 640°C. This suggests that the Stirling quartzites contain multiple components of magnetization that reside in hematite and that cannot be separated by thermal demagnetization. These negative results discouraged us from sampling the quartzitic B member.

A thin section of pale reddish quartzite from the upper C member of the Stirling, representative of the quartzites with scattered but stable directions, was examined in transmitted and reflected light. Discrete grains of specular hematite comprise less than 0.5% of the rock. They are commonly squarish, which suggests they may be martite; in addition, they show rounding and crisp contacts, which suggest they are detrital. Aggregates containing earthy and/or fine-grained specular hematite are also common. These aggregates apparently represent alteration 'ghosts' of iron-bearing minerals. Veins containing earthy and specular hematite also occur, and an orange cast pervades the rock.

The thin section observations may partly explain the magnetic behavior of the Stirling quartzites. Several distinct forms of hematite are present, and all probably contribute in varying degrees to the observed remanence. *Roy and Robertson* [1978] suggested that scattered, stable directions can arise from the presence of multiple components of magnetization, perhaps acquired during a time of frequent polarity reversals. Several components of magnetization could be present in the various froms of hematite observed in the Stirling, and this may account for the scattered directions of magnetization.

Discussion

The paleomagnetic data presented above suggest that both the Rainstorm Member of the Johnnie Formation and the C member of the Stirling Quartzite contain characteristic magnetizations acquired penecontemporaneously with deposition in the latest Precambrian. The precise direction of this characteristic magnetization could be determined only in the Rainstorm Member, in which two polarities of the magnetization are present. The corresponding north paleomagnetic pole from the Rainstorm is displaced about 47° from the north paleomagnetic pole from the stratigraphically higher Wood Canyon Formation of Early Cambrian age.

For purposes of paleogeographic reconstruction and stratigraphic correlation it is important to place constraints on the timing of this 47° apparent polar shift. The absolute age of the Rainstorm Member is unknown because of the absence of any direct isotopic age control in the late Precambrian miogeoclinal sequence in the southern Great Basin. Estimates of the absolute age of the Rainstorm can be made from sedimentation rates derived for the strata of the Desert Range. Early through Late Cambrian strata composing the uppermost Stirling Quartzite and overlying Wood Canyon, Carrara, Bonanza King, and Nopah Formations total about 3000 m. The dura-

TABLE 2. Paleomagnetic Poles From Laurentia That May Date From 675 to 575 m.y. B.P.

Pole	Pole Position	Rock Unit and Location	Age	Reference
1	37°N, 122°E	Intrusives, Colorado	Late Precambrian to Early Ordovician (704–485 m.y. B.P.)	*Larson and Mutschler* [1971], *French et al.* [1977]
2	44°N, 100°E			
3	15°N, 142°E			
4	5°N, 174°E			
5	48°N, 107°E			
6	5°N, 172°E	Cloud Mountain Basalt, western Newfoundland	Late Precambrian to Early Cambrian (605 ± 10 m.y. B.P.)	*Deutsch and Rao* [1977]
7	4°S, 161°E	Franklin rocks, Canadian Arctic (Group A)	Late Precambrian (675–625 m.y. B.P.)	*Palmer and Hayatsu* [1975]
8	8°N, 166°E	Franklin rocks, Canadian Arctic (Group B)		
9	4°N, 199°E*	Tillite Formation, East Greenland	~600 m.y. B.P.	*Bidgood and Harland* [1961]
10	13°S, 219°E*	Multicoloured Series, East Greenland	~600–800 m.y. B.P.	
11	9°S, 159°E	Mafic rock, Michigan	Late Precambrian to Cambrian(?)	*Van der Voo and Watts* [1978]
12	4°N, 169°E	Upper Keweenawan overprint, Michigan	Late Precambrian to Cambrian(?)	*Henry et al.* [1977]
13	17°N, 137°E	Tudor Gabbro, Ontario	(1200–670 m.y. B.P.)	*Palmer and Carmichael* [1973], *Hayatsu and Palmer* [1975]

*Pole uncorrected for rotation of Greenland with respect to North America.

tion of the Cambrian has been estimated to be about 75 m.y. [*Lambert*, 1971]. On this basis the average deposition rate of Cambrian strata in the Desert Range is roughly 40 m/m.y. Essentially the same rate is obtained by arbitrarily dividing the Cambrian into thirds and computing the deposition rate from the Early Cambrian strata alone. This suggests that the change from dominantly terrigenous sedimentation in the late Precambrian and Early Cambrian was not accompanied by a gross change in deposition rate. From the faunal evidence summarized earlier, the Precambrian-Cambrian boundary probably lies in the upper Stirling Quartzite [*Nelson*, 1978]. The pole from the Rainstorm Member of the Johnnie comes from beds approximately 800 m below the suggested Precambrian-Cambrian boundary. If the late Precambrian strata in the Desert Range were deposited at the rate of 40 m/m.y., estimated for the Cambrian, then the Rainstorm would be approximately 20 m.y. older than the Precambrian-Cambrian boundary. Assuming this boundary to have an age of about 590 m.y. B.P. [*Lambert*, 1971], the age of the Rainstorm would be about 610 m.y. B.P.

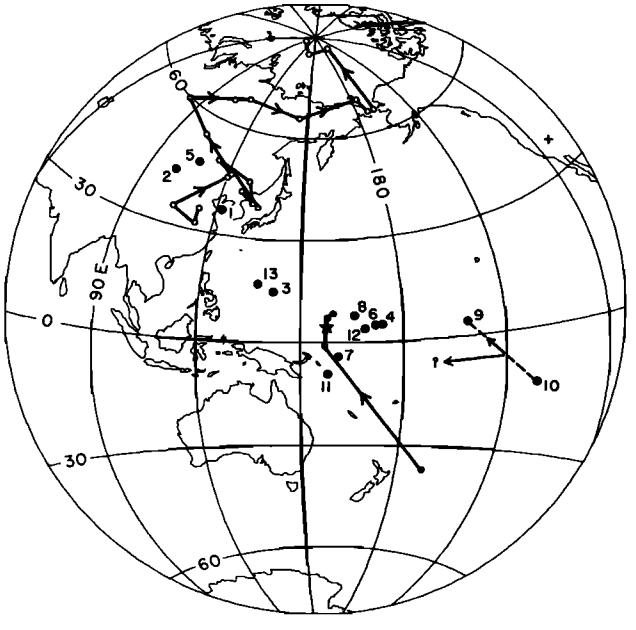


Fig. 8. Paleomagnetic poles from North America and Greenland that may represent the time 675–575 m.y. B.P. Pole numbers are as in Table 2. Also shown are the late Precambrian and Cambrian apparent polar wander path derived from the Desert Range (this study and *Gillett and Van Alstine* [1979]) and the post-Early-Ordovician part of the apparent polar wander path for North America [*Van Alstine*, 1979].

A second very rough estimate of the absolute age of the Rainstorm can be made from consideration of sedimentation rates off the Gulf Coast of the United States. In this region, thick near-shore accumulations of sediments are thought to represent a modern analog of miogeoclinal sedimentation [*Matthews*, 1974, pp. 82–88]. During the Cenozoic, a time comparable in length to the Cambrian, miogeoclinal sediments were deposited at rates ranging between 20 m/m.y. off Florida to 200 m/m.y. near the Mississippi delta [*Matthews*, 1974, p. 88]. If these limits bracket the deposition rates of the late Precambrian strata of the Desert Range, then the Rainstorm would lie between 4 and 40 m.y. below the Precambrian-Cambrian boundary. Because the Precambrian-Cambrian boundary probably lies in the range 610–570 m.y. B.P. [*Lambert*, 1971], reasonable limits to the absolute age of the Rainstorm would be about 650–575 m.y. B.P.

A third estimate of the age of the Rainstorm can be made on the basis of lithologic correlations and isotopic age control from other parts of the Cordilleran Geosyncline. *Stewart* [1972] has emphasized the stratigraphic similarities of the belt of upper Precambrian to Lower Cambrian strata extending from Alaska to northern Mexico. The basal units of these

successions characteristically contain diamictite (possibly of glacial origin) and tholeiitic basalt, overlain by thick sections of siltstone, shale, argillite, quartzite, and conglomerate. In the southern Great Basin the diamictite and associated volcanics occur in the Kingston Peak Formation, which unconformably underlies the Noonday Dolomite; the Noonday is conformable with the overlying Johnnie Formation. Stewart proposed that the diamictite and associated volcanics mark the initiation of a late Precambrian continental separation, followed by deposition of a thick wedge of miogeoclinal sediments on a trailing edge continental margin. This separation must be younger than the youngest rocks of the Belt Supergroup (about 850 m.y. B.P.; cf. *Stewart* [1972]). Assuming (1) an 850 m.y. B.P. age for the base of the Noonday Dolomite, (2) a constant sedimentation rate for the overlying late Precambrian strata, and (3) a 590 m.y. B.P. age for the Precambrian-Cambrian boundary, the Rainstorm would have an age of about 675 m.y. B.P. This estimate is based on the thickness of the late Precambrian strata in the Nopah Range, where a complete sequence is exposed from the basal Noonday Dolomite to the Middle Cambrian Bonanza King Formation. The three assumptions made in this calculation, however, would indicate an average sedimentation rate of 7 m/m.y. in the Nopah Range, which seems rather slow for miogeoclinal sedimentation. Hence the 675 m.y. B.P. estimate probably represents a maximum age for the Rainstorm.

The maximum and minimum age estimates set forth above are 675 and 575 m.y. B.P. We suggest that the age of the Rainstorm Member of the Johnnie Formation is about 625 ± 50 m.y. B.P.

Few paleomagnetic poles have been reported from rocks of Laurentia (i.e., North America, Greenland, and the Lewisian Platform of northwest Scotland) that might date from 675 to 575 m.y. B.P. (Table 2 and Figure 8). Colorado intrusives studied by *French et al.* [1977] and by *Larson and Mutschler* [1971] are dated as 704–485 m.y. B.P. and give a multimodal grouping of directions (represented by poles 1–5 of Figure 8). Pole 6, from three flows of the late Precambrian to Early Cambrian (605(?) m.y. B.P.) Cloud Mountain Basalts, western Newfoundland, may have incompletely averaged secular variation; in addition, this pole may reflect a vertical-axis rotation of that island [*Deutsch and Rao,* 1977].

Perhaps the most reliable late Precambrian poles for Laurentia come from flows, dikes, and sills of the Canadian Arctic [*Fahrig et al.,* 1971; *Robertson and Baragar,* 1972; *Fahrig and Schwarz,* 1973; *Palmer and Hayatsu,* 1975]. *Palmer and Hayatsu* [1975] have shown that lavas of the Natkusiak Formation and apparently related diabase sills from Victoria Island have a single-polarity (reversed?) magnetization with a pole (pole 7) similar to that from the Coronation sills of the mainland. The Natkusiak Formation disconformably overlies the upper Precambrian Shaler Group and is disconformably overlain by Cambrian strata [*Fahrig et al.,* 1971]. K-Ar age determinations on two sites in the least altered lavas gave a whole-rock isochron at 625 m.y. B.P. In contrast to these single-polarity results, possibly coeval intrusives of the central and eastern Arctic yield two-polarity magnetizations with a slightly more northerly pole position (pole 8). The mean of post-1963 K-Ar age determinations on these intrusives is in agreement with the isochron from the Natkusiak [*Palmer and Hayatsu,* 1975; *Fahrig et al.,* 1971]. However, Fahrig et al. preferred an age of 675 m.y. for the intrusives from the central and eastern Arctic.

Other latest Precambrian to earliest Cambrian poles come from reconnaissance paleomagnetic sampling by *Bidgood and Harland* [1961] of the thick, generally comformable succession of East Greenland. Pole 9, from the Tillite Group of latest Precambrian or earliest Cambrian age, is derived from the 16 most stable samples. Although no demagnetization studies were performed on these rocks, stability of the magnetization is suggested by the reversal and fold tests. In addition, 10 samples from the Multicoloured Series of the upper Precambrian Upper Eleonore Bay Group gave a single-polarity (reversed?) direction with a pole (pole 10) similar to that from the Tillite Group; however, no field or laboratory tests for stability were reported. The Multicoloured Series underlies the Limestone-Dolomite Series, which contains stromatolites comparable to those from Spitsbergen assigned a Vendian age (700–600 m.y. B.P.) [*Henriksen and Higgins,* 1976]. If the poles from East Greenland were corrected for rotation of Greenland with respect to North America, they would become more consistent with the late Precambrian and Early Cambrian results from elsewhere in Laurentia.

Several paleomagnetic poles have recently been reported from upper Precambrian or Lower Cambrian rocks of the Lake Superior region. Most of these poles are from the poorly dated Jacobsville Formation [*French,* 1976; *Roy and Robertson,* 1978], for which estimated ages range from late Precambrian to Triassic [*Hamblin,* 1958]. The most thorough paleomagnetic study of the Jacobsville [*Roy and Robertson,* 1978] supports an age greater than 1000 m.y. B.P., consistent with late Precambrian ages previously suggested for the Jacobsville by *Dubois* [1962] and *Babcock* [1975]. *Van der Voo and Watts* [1978] reported a paleomagnetic pole (pole 11) similar to that of the Franklin Group A pole from a deep (5.32 km), fully oriented core in a mafic unit. The regional stratigraphy suggests that this unit is an altered Keweenawan lava, but on the basis of its pole position these authors suggest that it was completely remagnetized in the late Precambrian to Cambrian. Finally, *Henry et al.* [1977] have reported a secondary magnetization in the Nonesuch Shale (~1060 m.y. B.P.) and Freda Sandstone (~1050 m.y. B.P.); they suggested this magnetization might have been acquired during an episode of late Precambrian copper mineralization. However, this secondary magnetization, represented by pole 12, also may well be older than 675 m.y. B.P.

Pole 13, from the Tudor Gabbro in the Grenville Province [*Palmer and Carmichael,* 1973], may date from about 670 m.y. B.P. Although Palmer and Carmichael thought this gabbro to have been emplaced at about 1200 m.y. B.P. (on the basis of isotopic ages on possibly coeval intrusives), no isotopic ages had been obtained from the Tudor itself. *Hayatsu and Palmer* [1975] have since reported a 670-m.y. K-Ar isochron for the Tudor Gabbro. However, the relation of this K-Ar age to the age of the magnetization is unclear. In addition, a controversy exists about whether the Grenville Province has undergone significant rotation with respect to the rest of North America [e.g., *Irving and McGlynn,* 1976].

None of the late Precambrian poles that might be younger than 675 m.y. B.P. is close to the pole from the Rainstorm Member of the Johnnie Formation (Figure 8). This suggests that the Rainstorm pole is not contemporaneous with any of them, a hypothesis permitted by the imprecise absolute ages for all these poles.

Morris and Roy [1977] and *Roy and Robertson* [1978] presented evidence for a nearly 180° apparent polar shift with respect to North America between about 1000 and 675 m.y.

B.P. The shift subsequent to 675 m.y. B.P. inferred from paleomagnetic investigations in the Desert Range may represent the later part of a long history of large-amplitude apparent polar wandering in the late Precambrian.

CONCLUSION

The 47° difference between the paleomagnetic pole from the uppermost Johnnie Formation (late Precambrian; 625 ± 50 m.y. B.P.) and the pole from the middle Wood Canyon Formation (Early Cambrian) suggests that at least 45° of apparent polar wandering occurred with respect to North America in latest Precambrian time.

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